

Install WSL (Skip if you are already using Ubuntu)

```
wsl --install
```

X Server (Optional. Only needed for GUI.)

download and install <https://sourceforge.net/projects/vcxsrv/>
Start the X server application.

Install QEMU

```
sudo apt update  
sudo apt upgrade  
sudo apt install qemu-system qemu-utils  
sudo apt install flex bison gawk qemu-system-x86 xorriso grub-common  
grub-pc-bin
```

Build XINU OS

```
git clone https://github.com/zrafa/xinu-x86-gui
```

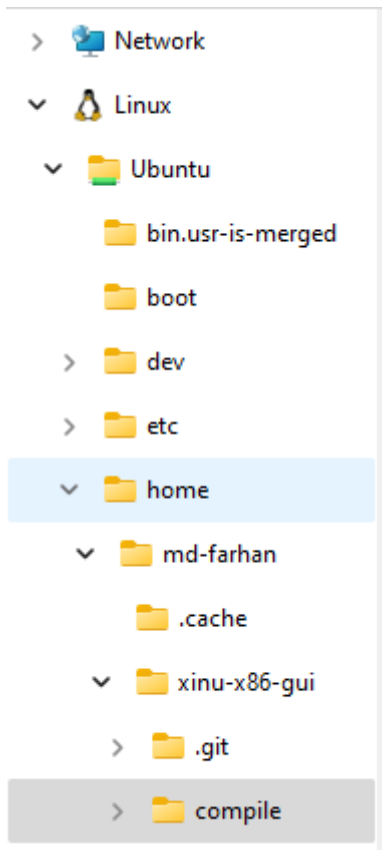
Navigate to the [Makefile](#) using file explorer.

For WSL.

```
Linux > Ubuntu > home > username > xinu-x86-gui > compile
```

For Ubuntu

```
home > username > xinu-x86-gui > compile
```



Inside `Makefile`, find `run-qemu` and add this to the end of the command `-nographic`

```
172 # Generates the Grub2-based ISO - see bochssrc.txt for info.
173 # Requires grub2-mkrescue and xorriso be installed!
174 run-qemu: qemu-iso-pre
175     echo OTRO GRUB PARA PROBAR: grub-mkrescue -o disk.iso test/img
176     grub-mkrescue -d /usr/lib/grub/i386-pc -o disk.iso test/img
177     qemu-system-i386 -machine accel=kvm -enable-kvm -drive
        file=disk.iso,index=0,media=cdrom -netdev user,id=u0 -device
        e1000-82545em,netdev=u0 -nographic
```

(run once for every time you boot WSL/Ubuntu)

```
sudo chmod 666 /dev/kvm
```

(run every time you boot XINU)

```
cd xinu-x86-gui/compile
```

```
make clean
```

```
make
```

```
make run-qemu
```

*Xinu OS GUI: vga, kbd and mouse

Use the ↑ and ↓ keys to select which entry is highlighted.
Press enter to boot the selected OS, 'e' to edit the commands
before booting or 'c' for a command-line.
The highlighted entry will be executed automatically in 3s.

XINU

Welcome to Xinu!

type help [ENTER] for ehhhh... help.

xsh \$ Obtained IP address 10.0.2.15 (0x0a00020f)

Thu Jan 9 8:07:46 EST 2025

mouse x: 0, y: 0

xsh \$

Shut down XINU

Open another terminal. (For WSL, you need to start a new WSL terminal.)

Use the **top** command to see running processes. Get PID of QEMU.

Press q to exit top.

kill <PID>

```
md-farhan@DESKTOP-SJGL3N  X  md-farhan@DESKTOP-SJGL3N  X  +  v
md-farhan@DESKTOP-SJGL3N:~$ top
top - 19:11:52 up 1:19, 1 user, load average: 0.93, 1.22, 0.98
Tasks: 32 total, 1 running, 31 sleeping, 0 stopped, 0 zombie
%Cpu(s): 33.3 us, 42.4 sy, 0.0 ni, 15.2 id, 3.0 wa, 0.0 hi, 6.1 si, 0.0 st
MiB Mem : 1875.0 total, 75.5 free, 540.4 used, 1438.3 buff/cache
MiB Swap: 1024.0 total, 1021.7 free, 2.3 used, 1334.6 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
 12799 md-farh+  20   0 542432 72200 28956 S  76.9   3.8   0:16.52 qemu-system-i386
    1 root      20   0  22228 13292  9520 S   0.0   0.7   0:09.00 systemd
    2 root      20   0   2616  1420  1320 S   0.0   0.1   0:00.07 init-systemd(Ub
    6 root      20   0   2644   132   132 S   0.0   0.0   0:01.41 init
   60 root      19  -1  66832 15780 14528 S   0.0   0.8   1:07.32 systemd-journal
  107 root      20   0  24092  6084  4788 S   0.0   0.3   0:03.79 systemd-udev
  118 systemd+  20   0  21452 11696  9504 S   0.0   0.6   0:00.37 systemd-resolve
  119 systemd+  20   0  91020  6460  5612 S   0.0   0.3   0:00.56 systemd-timesyn
  159 root      20   0   4236  2668  2444 S   0.0   0.1   0:00.02 cron
  160 message+  20   0   9756  5324  4372 S   0.0   0.3   0:01.94 dbus-daemon
  170 root      20   0  17976  8352  7332 S   0.0   0.4   0:00.84 systemd-logind
  173 root      20   0 1681852 11256  4768 S   0.0   0.6   0:02.01 wsl-pro-service
  182 root      20   0   3160  1072   988 S   0.0   0.1   0:00.02agetty
  189 root      20   0   3116  1076   988 S   0.0   0.1   0:00.02agetty
  196 root      20   0 107000 22024 12696 S   0.0   1.1   0:00.31unattended-upgr
  288 root      20   0   2624    96     0 S   0.0   0.0   0:00.00SessionLeader
  289 root      20   0   2624   112     0 S   0.0   0.0   0:02.66Relay(292)
  292 md-farh+  20   0   6072  4464  3036 S   0.0   0.2   0:00.70bash
  293 root      20   0   6696  3888  3276 S   0.0   0.2   0:00.02login
  382 md-farh+  20   0  20256 11276  9196 S   0.0   0.6   0:00.68systemd
  383 md-farh+  20   0  21148   880     0 S   0.0   0.0   0:00.00(scd-pam)
  394 md-farh+  20   0   6056  3500  2000 S   0.0   0.2   0:00.03bash
   959 polkitd  20   0 308160  6768  6120 S   0.0   0.4   0:00.93polkitd
md-farhan@DESKTOP-SJGL3N:~$ kill 12799
```

Running code in XINU

Create a c file, `xsh_hello.c`, in `xinu-x86-gui > shell`

	xsh_uptime.c	09-Jan-25 6:53 PM	C File	2 KB
	xsh_hello.c	09-Jan-25 7:16 PM	C File	0 KB

Open and write code inside it

```
1  #include <xinu.h>
2  #include <stdio.h>
3
4  shellcmd xsh_hello(int nargs, char *args[]) {
5      printf("Hello World\n");
6      return 0;
7  }
8
```

Open `cmdtab.c` from same directory (`xinu-x86-gui > shell`)

	cmdtab.c	09-Jan-25 6:53 PM	C File	2 KB
	lexan.c	09-Jan-25 6:53 PM	C File	4 KB

Add new entry

```
3  #include <xinu.h>
4  #include "shprototypes.h"
5
6  /*****
7   * Table of Xinu shell commands and the function address
8   *****/
9  const struct cmdent cmdtab[] = {
10     {"hello", FALSE, xsh_hello},
11     {"argecho", TRUE, xsh_argecho},
12     {"arp", FALSE, xsh_arp},
13     {"cat", FALSE, xsh_cat},
```

(`FALSE`, because `hello` is not a built in command)

Open `shprototypes.h` header file from `xinu-x86-gui > include`

	shell.h	09-Jan-25 6:53 PM	H File	3 KB
	shprototypes.h	09-Jan-25 6:53 PM	H File	3 KB
	stdarg.h	09-Jan-25 6:53 PM	H File	1 KB

Add prototype of our function

```
1  /* in file xsh_hello.c */
2  extern shellcmd xsh_hello (int32, char *[]);
3
```

Open `process.h` header file from `xinu-x86-gui > include`

 <code>process.h</code>	10-Jan-25 7:25 PM	H File	3 KB
 <code>prototypes.h</code>	09-Jan-25 6:53 PM	H File	17 KB

Change `NPROC` from 8 to 20

```
#ifndef NPROC
#define NPROC      20
#endif
```

Build and run the the OS from `xinu-x86-gui > compile` directory

`cd xinu-x86-gui/compile`

`make clean`

`make`

`make run-qemu`

In xinu shell, type command `hello` and press enter

```
xsh $
xsh $
xsh $
xsh $ hello
Hello World
xsh $
```

Troubleshooting

This section was contributed by Syed Shifat E Rahman (2003065)

You might encounter grub related errors while running `make run-qemu`.

```
pc:~/xinu-x86-gui/compile$ make run-qemu
Rebuilding the .defs file

Loading object files to produce GRUB bootable xinu
/usr/bin/ld: warning: binaries/kbdispatch.o: missing .note.GNU-stack section implies executable stack
/usr/bin/ld: NOTE: This behaviour is deprecated and will be removed in a future version of the linker

XINU ready.

make run-qemu # ...to boot the OS on qemu

rm -rf test/img
mkdir -p test/img/boot/grub
cp xinu.elf test/img/
cp test/grub/grub.cfg test/img/boot/grub
echo OTRO GRUB PARA PROBAR: grub-mkrescue -o disk.iso test/img
OTRO GRUB PARA PROBAR: grub-mkrescue -o disk.iso test/img
grub-mkrescue -d /usr/lib/grub/i386-pc -o disk.iso test/img
grub-mkrescue: error: /usr/lib/grub/i386-pc/modinfo.sh doesn't exist. Please specify --target
or --directory.
make: *** [Makefile:176: run-qemu] Error 1
```

```
sudo apt install grub-pc-bin  
sudo apt reinstall grub-pc-bin  
grub-mkrescue --target=i386-pc -d /usr/lib/grub/i386-pc -o disk.iso test/img  
make run-qemu
```